Enforcing hard constraints in physics-informed learning for transient TSV electromigration analysis

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ABSTRACT

Due to the high local current densities, Through Silicon Vias (TSVs) are susceptible to electromigration (EM) degradation, which reduces the reliability of integrated circuits. Unlike traditional methods for TSV modeling and simulation, this paper introduces a unified hard constraint physics-informed learning neural network approach, called HCPINN, for the transient analysis of electromigration in TSVs by solving the conventional mass balance equation. The proposed method allows simultaneous consideration of atomic depletion and accumulation, effective resistance degradation, electric current evolution, and stress distribution. Specifically, we propose a hard constraint method for solving partial differential equations (PDEs) with general boundary conditions (BCs) for transient TSV electromigration analysis. By using the extra fields derived from the mixed finite element method, we reconstruct the corresponding PDEs by transforming general BCs into linear forms. Based on this derivation, we embed general BCs of mass balance equation into the proposed ansatz and employ sub-networks for the approximation on general BCs. The main neural network is responsible for training the internal part of the problem domain without adding loss terms with BCs, overcoming the convergence issue due to unbalanced gradients among different loss terms. Besides, we theoretically demonstrate that this reformulation of general BCs can stabilize the training process. Experimental results indicate that the proposed HCPINN exhibits superior performance and reduces boundary error in TSV electromigration analysis. Compared to the finite element method, the proposed network achieves approximately 100 times faster inference with a minimal mean squared error increase of less than 0.1%.

1 INTRODUCTION

3-D integrated circuits (ICs) that leverage stacked dies and through-silicon vias (TSVs) have recently received a lot of attention because of its potential to break through the limit of traditional CMOS scaling, to facilitate system-in-package and to advance the "More than Moore" scaling principle into the third dimension. However, the reliability of TSV-based 3-D IC is limited by electromigration, which even potentially leads to diminished efficiency and the risk of short or open circuit failures. One of the factors that pose a threat to reliability of TSVs is the carrying intense currents and high local current densities in TSVs due to current crowding [1], as regions of high local current densities are more vulnerable to electromigration degradation. Additionally, the mechanical and thermal stresses that the vertical interconnect with TSV bears also have great influence on electromigration (EM) failure. Considering the broad application prospects of 3D IC technology, it is necessary to comprehensively understand and address the EM reliability issue for TSV-based 3-D ICs.

Early research for EM reliability of TSV mainly focused on thermomechanical stress simulation and evaluation by finite element method (FEM) [2], including the evaluation of the fatigue of Cu TSVs, delamination at the TSV interfaces, and silicon cracking induced by thermomechanical stresses. Most works on TSV based EM analysis focused on the diffusion model of mass flux on several faces of TSVs. Numerous numerical experiments have been conducted to characterize the electrical, stress, thermal, and atomic concentration fields within TSV structures [3, 4]. Pak et al. [5] investigated EM lifetime modeling for TSVs, focusing particularly on the impact of TSV-induced stress on the EM lifetimes of Back-End-of-Line (BEOL) wires. Advancing these studies, Zhao et al. [6] explored time-dependent EM modeling specific to TSV-based 3D connections, opting to consider the grain structure in their EM assessments rather than modeling the entire TSV as a single crystal. Additionally, Zhou et al. [7] examined and modeled thermal stresses, employing these findings in the reliability analysis of TSVs. Despite these advancements, the numerical methods utilized in these studies are often limited by the necessity for fine grid meshing and temporal iterations, resulting in significant computational costs particularly when striving to achieve accurate and self-consistent solutions to partial differential equations (PDEs).

Machine learning (ML) methods [8], especially physics-informed neural networks (PINNs) [9], have demonstrated their capability to address the challenges associated with solving PDEs. A key feature of PINNs is the application of automatic differentiation (AD) technology [10], which facilitates the computation of derivative terms and enhances the computation efficiency compared with traditional numerical methods predominantly depending on discretization techniques. ML and PINN-based methods have been applied for EM analysis in many recent studies. Hou et al. [11] developed a space-time physics-informed neural network (STPINN) for computing EM stress evolution governed by Korhonen's model along multi-segment interconnect trees during void nucleation phase under

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various thermal conditions. Jin et al. [12] implemented HierPINN-EM to mitigate the slow training and convergence issues associated with standard PINN models. Additionally, Hou et al. [13] utilized multilayer perceptrons (MLPs) for obtaining stress evolution within interconnect trees throughout the void nucleation phase. An innovative strategy, PostPINN-EM, was introduced by Lamichhane [14] for addressing the partial differential equations governing stress evolution for post-void phase in confined metal interconnect multi-segment trees.

Theoretical researches [15, 16] have shown that loss terms associated with PDEs and BCs have unbalanced convergence rate during training process, leading to the total training error. To address this problem, some researchers [17–19] have tried to embed BCs into the network structure directly and avoid adding loss component of BCs. Some works [20–23] transformed the initial problem into the reformulated weak form or the variational form of the PDEs, thus lowering the smoothness requirement and potentially alleviating the computation cost during optimization. Besides, the Theory of Connections proposed in [24] has been adopted for overcoming the issue of unbalanced convergence rate [25, 26]. However, it should be noted that these methods are limited to specific boundary conditions such as Dirichlet BCs and homogeneous BCs. In order to solve PDEs with Neumann and Robin BCs, Liu et al. [27] introduced the "$extra\ fields$" from the mixed finite element method to reformulate the PDEs and transform the Dirichlet, Neumann and Robin BCs into linear forms, thereby deriving the analytical solutions of these BCs. One of the limitations of this method is that they only consider the three classical BCs and not applicable for more general BCs. Since the BCs of the mass balance equation governing TSV EM process contain more general BCs and cannot be solved by these above mentioned methods, we consider proposing a hard-constraint method extending the application for more general BCs.

In this paper, we propose a unified hard constraint physics-informed learning neural network, HCPINN, for solving mass balance equation and atomic flux diversity equations with general BCs in EM analysis of TSV. With the proposed HCPINN framework, we are able to substitutes the gradient of atomic concentration and the product of this gradient with new variables and construct an ansatz specific to TSV EM analysis. Unlike conventional methods, the proposed approach can eliminate grid meshing and temporal iteration requirements, which can significantly reduce computation cost and improve efficiency. Additionally, our approach exhibits promising accuracy, positioning it as a potentially superior alternative for computational electromigration modeling and TSV analysis. The main contributions are as follows:

- We propose a physics-informed learning method to solve the traditional mass balance equation in 3-D axis-symmetric geometry and to predict the evolution of atomic concentration, stress distribution, and electric current during electromigration process in TSV. To the best of our knowledge, this is the $first\ study$ that utilizes the learning capabilities of physics-informed neural network for multiphysics analysis of EM modeling in TSV.
- We improve the "$extra\ fields$" method presented in [27] for deriving the general solution of the more general BCs in EM analysis of TSV, and theoretically prove that the proposed method can stabilize the optimization process.
- We develop a customized unified hard constraint physics-informed neural network for simulating the EM process in TSV. The main neural network framework is designed to optimize the solution of internal problem domain, while other sub-networks are responsible for the approximation on general BCs in TSV EM analysis.
- The proposed method has superior performance over state-of-the-art baselines on solving PDEs in modeling and simulation of TSV electromigration process. Experimental results show that the proposed method can achieve more than 100 speedup compared to FEM based tool COMSOL with less than 0.1% accuracy.

This paper is organized as follows: Section 2 introduce the transient EM modeling governed by mass balance equation in TSV, including stress modeling and resistivity modeling. In Section 3, we propose a hard constraint method for solving PDEs with general BCs in TSV EM analysis, construct a novel neural network, HCPINN, for predicting the multi-physics variables, and theoretically demonstrate the effectiveness of the proposed method on improving training stability. Experimental results are detailed in Section 4. Finally, Section 5 concludes this paper.

2 TRANSIENT EM MODELING IN TSV

2.1 Fundamental PDEs

In this section, we explore transient electromigration modeling in TSV. The failure of TSV due to electromigration can result from several interacting forces, including current density, temperature gradient, mechanical stress gradient, and atom concentration gradient. To model the transient evolution of atomic concentration, represented by N, at any given point (x, y, z) and time t, we employ the conventional mass balance equation for EM modeling and simulation in TSVs, given by

$$\frac{\partial N}{\partial t} + \nabla \cdot (\mathbf{J_C} + \mathbf{J_S} + \mathbf{J_T} + \mathbf{J_N}) = 0, \tag{1}$$

where $\mathbf{J_C} = \frac{N}{kT} eZ^* \rho \mathbf{j} D$ represents the current-induced atomic flux propelled by electrical forces, $\mathbf{J_S} = \frac{N\Omega}{kT} D\ \nabla\sigma_H$ depicts the stress-induced atomic flux driven by mechanical stresses within the material, $\mathbf{J_T} = -\frac{NQ^*}{kT^2} D\ \nabla T$ is the temperature-gradient-induced atomic flux arising from thermal gradients, and $\mathbf{J_N} = -D\ \nabla N$ is the atomic flux due to atom diffusion, a consequence of the natural tendency of atoms to move from regions of high to low concentration. Here, D_0 is self-diffusion coefficient, E_A is activation energy, k is Boltzmann constant, T is temperature, e is electric charge, Z^* is effective valence charge, ρ is electrical resistivity, Q^* is heat of transport, Ω is atomic volume, $\mathbf{j}$ is current density, σ_H is hydrostatic stress, and $D = D_0 \exp(\frac{-E_A}{kT})$ is the effective atomic diffusion coefficient. Experiment results in [6] indicate that temperature gradients of copper TSV are very small due to the high thermal conductivity of copper. Consequently, $\mathbf{J_T}$ induced by temperature gradients is not considered in the proposed model, and a constant temperature T of 350K is assumed for the simulation.

2.2 Stress Modeling

A TSV-based 3D connection can be influenced by the back flow stress, thermal residual stress, and thermal stress. Back flow stress refers to the stress induced by the redistribution [28] of atoms within the via due to the passage of electrical current. This redistribution results in a variation in the local atomic concentration,

leading to the generation of mechanical stress within the material. The tensile stress is generated by the atomic depletion, while the compression stress results from atomic accumulation. Moreover, back flow stress is co-simulated with EM process and will increase over time due to EM-induced atomic concentration. Thermal residual stress arises from the variation of temperature during the manufacturing process [29]. For instance, the thermomechanical stress is generated during the fabrication process resulting from the mismatch of thermal expansions coefficients among different materials. The thermal residual stress may dissipate gradually [30], for which our EM modeling process ignores this kind of stress [6]. Thermal stress arises from the coefficients of thermal expansion (CTE) mismatch among different materials and the presence of elevated thermal gradients. In TSVs, the passage of electrical current induces local heating, leading to thermal expansion or contraction of the material and consequently generating thermal stress.

It is important to note that the stress-induced atomic flux, denoted as $\mathbf{J_S}$ in (1), is influenced by the gradient of hydrostatic stress σ_H. The hydrostatic stress primarily includes the back flow stress defined as $\sigma_{\text{bf}} = B\frac{N-N_0}{N_0}$ where B is the back flow stress modulus. Experimental results indicate that thermal expansion stress has a minimal impact on the overall electromigration atomic transport [6]. Consequently, we focus exclusively on back flow stress as the critical component of hydrostatic stress in our analysis.

2.3 Resistivity Modeling

Electromigration in TSVs can lead to metal ion accumulation and depletion at specific points, causing significant structural changes. According to [31], the changes include the formation of voids at the cathode end and hillocks or whiskers at the anode end, which alter the cross-sectional area available for current flow and thus impact the electrical resistance. The formation of voids reduces the effective cross-sectional area, increasing electrical resistance, while hillocks can decrease resistance, and even may lead to short circuits if they contact adjacent structures. Due to metal barrier layers, open circuit failures from voids are more common than short circuit failures from hillocks. Therefore, our analysis primarily focuses on void-related failures in TSVs. For simplicity, we adopt the approach from [6] that models TSV resistivity as a function of atomic concentration to estimate resistance. In our work, we propose that TSV will experience void formation if the atomic concentration falls below $0.85N_0$. Conversely, we assume that the resistivity of TSV to be ρ_0 when the atomic concentration exceeds $0.9N_0$. Based on resistivity measurements in [6], we assign a resistivity value of $32\rho_0$ to voids and apply a linear approximation for resistivity values between $0.85N_0$ and $0.9N_0$. Thus, the simplified model for TSV resistivity is presented as follows:

$$\rho = \begin{cases} 32\rho_0, \ 0 < N < 0.85N_0, \\ 32\rho_0 - 620 \times (\frac{N}{N_0} - 0.85)\rho_0, \ 0.85N_0 \le N \le 0.9N_0, \\ \rho_0, \ 0.9N_0 < N < 5N_0. \end{cases} \quad (2)$$

3 THE PROPOSED HCPINN METHOD

3.1 Problem Formation

For the atomic concentration evolution equation (1), we transform the 3-D model into 2-D model by utilizing the axisymmetric property of the proposed TSV structure, as shown in Figure 1. This conversion involves mapping from the Cartesian to the Cylindrical coordinate system, given by:

$$(\nabla \cdot \Gamma)_{(x,y,z)\to(r,z)} = \frac{1}{r}\frac{\partial(r\Gamma_r)}{\partial r} + \frac{\partial \Gamma_z}{\partial z}. \quad (3)$$

The radial coordinate of the axis of TSV is denoted as $r = 0$ and the radius of TSV as R_{tsv}, respectively. The height coordinates of the top electrodes (TE) and bottom electrodes (BE) are set to be $z = h$ and $z = 0$, respectively. Based on Equations (1) and (3), the evolution of atomic concentration in the TSV model can be rewritten as follows:

$$\frac{\partial N}{\partial t} + \nabla \cdot (-D\nabla N + \mathbf{A_{1r}}DN + \mathbf{A_{1z}}DN + A_2 DN\nabla N) = 0, \quad (4)$$

where $\mathbf{A_{1r}} = \frac{eZ^*\rho \mathbf{j_r}}{kT}, \mathbf{A_{1z}} = \frac{eZ^*\rho \mathbf{j_z}}{kT}, A_2 = \frac{\Omega B}{kTN_0}$, and $r \in [0, R_{tsv}]$, $z \in [0, h]$, $t > 0$. It should be noted that $\mathbf{j_r}$ and $\mathbf{j_z}$ are the current densities of r and z coordinates, respectively, and the divergence of flux $\mathbf{J} = \mathbf{J_C} + \mathbf{J_S} + \mathbf{J_N}$ in (1) is expected to be zero on the vertical and radial boundaries. Let $\mathbf{J_r} = -D\nabla N + \mathbf{A_{1r}}DN + A_2 DN\nabla N$, and $\mathbf{J_z} = -D\nabla N + \mathbf{A_{1z}}DN + A_2 DN\nabla N$, and thus the boundary conditions corresponding to Equation (4) can be expressed as follows:

$$(\mathbf{n} \cdot \mathbf{J_r})\big|_{r=R_{tsv}} = 0, \ (\mathbf{n} \cdot \mathbf{J_z})\big|_{z=0,h} = 0. \quad (5)$$

The corresponding initial atomic concentration within TSV is set as $N(r, z, 0) = N_0$. To obtain the current densities along r and z coordinates, we need to derive the following PDEs describing electric current field:

$$\nabla \cdot (-\sigma \nabla V(r, z, t)) = 0, \quad (6)$$

$$V(r, z, t)\big|_{z=0} = 0, \ V(r, z, t)\big|_{z=h} = \psi, \quad (7)$$

$$\mathbf{n} \cdot (-\sigma \nabla V)\big|_{r=R_{tsv}} = 0, \quad (8)$$

where $\sigma = 1/\rho$ is the conductivity of TSV in which ρ is calculated by Equation (2). Equation (7) indicates that the top electrodes serve as the electrical potential input ψ, while the bottom electrodes are ground. Equation (8) signifies electrical insulation on the boundary. Thus, the current densities $\mathbf{j_r}$ and $\mathbf{j_z}$ along r and z coordinates can be calculated by $\mathbf{j_r} = -\sigma V_r$ and $\mathbf{j_z} = -\sigma V_z$, respectively.

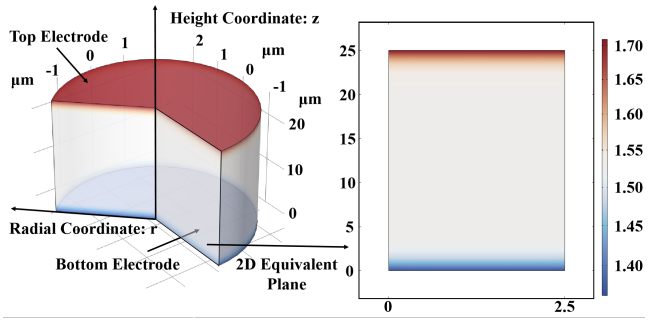


Figure 1: Converting the 3-D symmetrical physics-based model into 2-D model by mapping from the Cartesian to the Cylindrical coordinate system.

3.2 Hard Constraints

The boundary conditions can be embedded into the ansatz by using the hard constraint method. The instance from the ansatz can automatically satisfy the boundary conditions. Figure 2 illustrates the flow of the proposed hard constraint method. First, we abstract

the PDEs with general BCs in TSV EM analysis to the form:

$$\mathcal{F}[u(\boldsymbol{x})] = \mathbf{0},\ \boldsymbol{x} = (x_1, \ldots, x_d) \in \Omega, \tag{9}$$

where $\Omega \subset R^d$ is the problem domain , $\mathcal{F} = (\mathcal{F}_1, \ldots, \mathcal{F}_N)$ includes N operators which map u to a function of $\boldsymbol{x}$, u, derivatives of u and $u\nabla u$. It is noted that $u(\boldsymbol{x})$ is the solution to the original PDEs. Subsequently, we abstract the general BCs in mass balance equation and depict them as the following form:

$$a_i(\boldsymbol{x})u + b_i(\boldsymbol{x})(\boldsymbol{n}(\boldsymbol{x}) \cdot \nabla u) + c_i(\boldsymbol{x})(\boldsymbol{n}(\boldsymbol{x}) \cdot u\nabla u) = g_i(\boldsymbol{x}), \tag{10}$$

where $\boldsymbol{x} \in \gamma_i (i = 1, \ldots, n)$, γ_i is the subset of the boundary $\partial\Omega$, $a_i(\boldsymbol{x}), b_i(\boldsymbol{x})$ and $c_i(\boldsymbol{x})$ satisfy that $a_i(\boldsymbol{x})^2 + b_i(\boldsymbol{x})^2 + c_i(\boldsymbol{x})^2 \neq 0$ holds for $\forall \boldsymbol{x} \in \gamma_i$, and $\boldsymbol{n}(\boldsymbol{x}) = (n_1(\boldsymbol{x}), \ldots, n_d(\boldsymbol{x}))$ is the outward facing unit normal at $\boldsymbol{x} \in \gamma_i$. To derive the solutions of general BCs in Equation (10), we let $\boldsymbol{p}(\boldsymbol{x}) = \nabla u$, $\boldsymbol{q}(\boldsymbol{x}) = u\nabla u$, and reformulate the Equation (10) into the following linear forms:

$$a_i(\boldsymbol{x})u + b_i(\boldsymbol{x})\big(\boldsymbol{n}(\boldsymbol{x}) \cdot \boldsymbol{p}(\boldsymbol{x})\big) + c_i(\boldsymbol{x})(\boldsymbol{n}(\boldsymbol{x}) \cdot \boldsymbol{q}(\boldsymbol{x})) = g_i(\boldsymbol{x}). \tag{11}$$

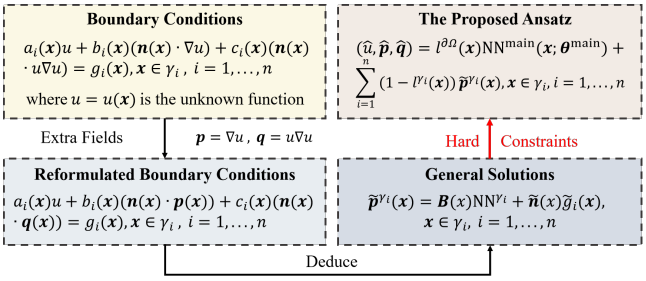


Figure 2: Flow of calculating the hard constraints and the ansatz.

The general BCs in TSV electromigration analysis is transformed into linear equations with respect to $(u, \boldsymbol{p}, \boldsymbol{q})$ by the hard constraint method, which are much easier to derive general solutions. For simplicity, we denote $(u, \boldsymbol{p}, \boldsymbol{q})$ by $\tilde{\boldsymbol{p}}$, and their general solutions at γ_i by $\tilde{\boldsymbol{p}}^{\gamma_i}$, respectively. Let $\tilde{\boldsymbol{n}} = (a_i, b_i\boldsymbol{n}, c_i\boldsymbol{n})/\sqrt{a_i^2 + b_i^2 + c_i^2}$, $\tilde{g}_i = g_i/\sqrt{a_i^2 + b_i^2 + c_i^2}$, and then Equation (11) is equivalent to

$$\tilde{\boldsymbol{n}}(\boldsymbol{x}) \cdot \tilde{\boldsymbol{p}}^{\gamma_i}(\boldsymbol{x}) = \tilde{g}_i(\boldsymbol{x}). \tag{12}$$

To obtain the general solution $\tilde{\boldsymbol{p}}^{\gamma_i}$, the first step is to find a basis $\boldsymbol{B}(\boldsymbol{x})$ of the null space whose dimension is $d' = 2d$. If $a_i(\boldsymbol{x}) = 0$, the value of d' will minus one; if $b_i(\boldsymbol{x}) = 0$ or $c_i(\boldsymbol{x}) = 0$, the value of d' will minus two. It should be noted that the columns of $\boldsymbol{B}(\boldsymbol{x})$ cannot degenerate into linearly vectors, otherwise $\boldsymbol{B}(\boldsymbol{x})$ will not to be able to represent all possible solutions of boundary conditions. Therefore, we choose $d' + 1$ vectors in the null space as $\boldsymbol{B}(\boldsymbol{x})$, d' of which are linearly independent. We perform a Gram-Schmidt orthogonalization of $\tilde{\boldsymbol{n}}$ on each vector in the standard basis to get $\boldsymbol{\beta}_k(\boldsymbol{x}) = \boldsymbol{e}_k - \frac{\boldsymbol{e}_k \cdot \tilde{\boldsymbol{n}}(\boldsymbol{x})}{\tilde{\boldsymbol{n}}(\boldsymbol{x}) \cdot \tilde{\boldsymbol{n}}(\boldsymbol{x})}\tilde{\boldsymbol{n}}(\boldsymbol{x}) = \boldsymbol{e}_k - (\boldsymbol{e}_k \cdot \tilde{\boldsymbol{n}}(\boldsymbol{x}))\,\tilde{\boldsymbol{n}}(\boldsymbol{x})$, $k = 1, \ldots, d' + 1$, where $[\boldsymbol{e}_1(\boldsymbol{x}), \ldots, \boldsymbol{e}_{d'+1}(\boldsymbol{x})] = \boldsymbol{I}_{d'+1}$. Thus, we set $\boldsymbol{B}(\boldsymbol{x}) = \big[\boldsymbol{\beta}_1(\boldsymbol{x}), \ldots, \boldsymbol{\beta}_{d'+1}(\boldsymbol{x})\big] = \boldsymbol{I}_{d'+1} - \tilde{\boldsymbol{n}}(\boldsymbol{x})\tilde{\boldsymbol{n}}(\boldsymbol{x})^\top$ and construct the ansatz as follows:

$$\tilde{\boldsymbol{p}}^{\gamma_i}(\boldsymbol{x}; \boldsymbol{\theta}^{\gamma_i}) = \boldsymbol{B}(\boldsymbol{x})\mathrm{NN}^{\gamma_i}(\boldsymbol{x}; \boldsymbol{\theta}^{\gamma_i}) + \tilde{\boldsymbol{n}}(\boldsymbol{x})\tilde{g}_i(\boldsymbol{x}), \tag{13}$$

where $\mathrm{NN}^{\gamma_i} : \mathbb{R}^{d'} \to \mathbb{R}^{d'+1}$ is a neural network with trainable parameters $\boldsymbol{\theta}^{\gamma_i}$. With the parameterization of NN^{γ_i}, Equation (13) can represent any function at γ_i as long as the function satisfies the BCs in Equation (10), which is theoretically demonstrated in the following theorem.

Theorem 1. *There exists $\boldsymbol{\theta}^{\gamma_i} \in \Theta^{\gamma_i}$ $(i = 1, \ldots, n)$ such that*

$$\left|\tilde{\boldsymbol{n}}(\boldsymbol{x}) \cdot \tilde{\boldsymbol{p}}^{\gamma_i}(\boldsymbol{x}) - \tilde{g}_i(\boldsymbol{x})\right| < \varepsilon,\ \forall \varepsilon > 0, \tag{14}$$

holds for all $\boldsymbol{x} \in \gamma_i$, where Θ^{γ_i} is the parameter space of the neural network NN^{γ_i}, $\tilde{\boldsymbol{n}}(\boldsymbol{x}) = (a_i, b_i\boldsymbol{n}(\boldsymbol{x}), c_i\boldsymbol{n}(\boldsymbol{x}))/\sqrt{a_i^2 + b_i^2 + c_i^2}$, and $\boldsymbol{B}(\boldsymbol{x}) = \boldsymbol{I}_{d'+1} - \tilde{\boldsymbol{n}}(\boldsymbol{x})\tilde{\boldsymbol{n}}(\boldsymbol{x})^\top$.

We need to combine the general solutions corresponding to each boundary γ_i for achieving an overall approximation. To achieve this, we propose the ansatz as follows

$$(\hat{u}, \hat{\boldsymbol{p}}, \hat{\boldsymbol{q}}) = l^{\partial\Omega}(\boldsymbol{x})\mathrm{NN}^{\mathrm{main}}(\boldsymbol{x}; \boldsymbol{\theta}^{\mathrm{main}}) + \sum_{i=1}^{n}(1 - l^{\gamma_i}(\boldsymbol{x}))\tilde{\boldsymbol{p}}^{\gamma_i}(\boldsymbol{x}; \boldsymbol{\theta}^{\gamma_i}), \tag{15}$$

where $\mathrm{NN}^{\mathrm{main}} : \mathbb{R}^{d'} \to \mathbb{R}^{d'+1}$ is the main neural network with trainable parameters $\boldsymbol{\theta}_{\mathrm{main}}$, $l^{\partial\Omega}$ and l^{γ_i} are extended distance functions, which are depicted as $l(\boldsymbol{x}) = 0$ if $\boldsymbol{x}$ is on the boundary, otherwise $l(\boldsymbol{x}) = 1$. The following theorem shows that the ground truth solution u to the PDEs can be approximated as $\hat{u}$ obtained by the proposed hard constraint method.

Theorem 2. *There exist $\boldsymbol{\theta}^{main} \in \Theta^{main}$ and $\boldsymbol{\theta}^{\gamma_i} \in \Theta^{\gamma_i} (i = 1, \cdots, n)$ such that $\|(\hat{u}, \hat{\boldsymbol{p}}, \hat{\boldsymbol{q}}) - (u, \nabla u, u\nabla u)\|_1 < \varepsilon$, holds for all $\varepsilon > 0$ and $\boldsymbol{x} \in \Omega \cup \partial\Omega$, where Θ is the parameter space of the corresponding neural network, $\|\cdot\|_1$ is the 1-norm, u is the ground truth solution to the PDEs.*

Theorem 1 demonstrates that the derived general solution, $\tilde{\boldsymbol{p}}^{\gamma_i}(\boldsymbol{x})$, satisfies the corresponding BC whose abstract form is presented in Equation (10). Theorem 2 states that the constructed ansatz is the approximation of the ground truth solution to related PDEs. Due to limited space, we omit the proofs of Theorems 1-2.

3.3 Physics-Informed Learning

In this section, we propose a physics-informed learning structure based on the hard constraint method for solving PDEs with general BCs for TSV EM modeling. First, we abstract BCs in Equation (5) and Equation (8) to this form : $a_i(\boldsymbol{x})u + b_i(\boldsymbol{x})(\boldsymbol{n}(\boldsymbol{x}) \cdot \nabla u) + c_i(\boldsymbol{x})(\boldsymbol{n}(\boldsymbol{x}) \cdot u\nabla u) = g_i(\boldsymbol{x})$, $\boldsymbol{x} = (r, z)$. For Equation (5), u represents atomic concentration N, $i = 1, 2$, the boundary of problem domain contains two subsets: $\gamma_1 : r = R_{tsv}$ and $\gamma_2 : z = 0 \cup z = h$, and $a_1(\boldsymbol{x}) = \mathbf{A_{1r}}D$, $a_2(\boldsymbol{x}) = \mathbf{A_{1z}}D$, $b_1(\boldsymbol{x}) = b_2(\boldsymbol{x}) = -D$, $c_1(\boldsymbol{x}) = c_2(\boldsymbol{x}) = A_2D$, $g_1(\boldsymbol{x}) = g_2(\boldsymbol{x}) = 0$. For Equation (8), u represents electric potential V, $i = 1$, and the boundary of problem domain is denoted as $\gamma_1 : r = R_{tsv}$, $a_1(\boldsymbol{x}) = 0$, $b_1(\boldsymbol{x}) = -\sigma$, $c_1(\boldsymbol{x}) = 0$, $g_1(\boldsymbol{x}) = 0$. Subsequently, we let $\boldsymbol{p}_N(\boldsymbol{x}) = \nabla N$, $\boldsymbol{p}_V(\boldsymbol{x}) = \nabla V$, $\boldsymbol{q}_N(\boldsymbol{x}) = N\nabla N$ to obtain the following reformulated form:

$$\begin{aligned}
&\mathrm{PDE}: \frac{\partial N}{\partial t} + \nabla \cdot (-D\boldsymbol{p}_N(\boldsymbol{x}, t) + \mathbf{A_{1r}}DN + \mathbf{A_{1z}}DN + A_2D\boldsymbol{q}_N(\boldsymbol{x}, t)) = 0,\\
&\qquad \nabla \cdot (-\sigma\boldsymbol{p}_V(\boldsymbol{x}, t)) = 0,\\
&\mathrm{Extra\ \ Fields}: \boldsymbol{p}_N(\boldsymbol{x}, t) = \nabla N,\ \boldsymbol{q}_N(\boldsymbol{x}, t) = N\nabla N,\ \boldsymbol{p}_V(\boldsymbol{x}, t) = \nabla V,\\
&\mathrm{BC}: \boldsymbol{n}(\boldsymbol{x}) \cdot (-D\boldsymbol{p}_N(\boldsymbol{x}, t) + \mathbf{A_{1r}}DN + A_2D\boldsymbol{q}_N(\boldsymbol{x}, t))\Big|_{\gamma_1} = 0,\\
&\qquad \boldsymbol{n}(\boldsymbol{x}) \cdot (-D\boldsymbol{p}_N(\boldsymbol{x}, t) + \mathbf{A_{1z}}DN + A_2D\boldsymbol{q}_N(\boldsymbol{x}, t))\Big|_{\gamma_2} = 0,\\
&\qquad \boldsymbol{n}(\boldsymbol{x}) \cdot (-\sigma\boldsymbol{p}_V(\boldsymbol{x}, t))\Big|_{\gamma_1} = 0,\ V(\boldsymbol{x}, t)\Big|_{\gamma_{21}} = 0,\ V(\boldsymbol{x}, t)\Big|_{\gamma_{22}} = \psi,\\
&\mathrm{IC}: N(\boldsymbol{x}, 0) = N_0,
\end{aligned} \tag{16}$$

where γ_{21} represents the boundary condition $z = 0$, γ_{22} represents the boundary condition $z = h$, and $\boldsymbol{p}_N(\boldsymbol{x}, t), \boldsymbol{p}_V(\boldsymbol{x}, t), \boldsymbol{q}_N(\boldsymbol{x}, t)$ are the introduced extra fields. Then, we let $\tilde{\boldsymbol{p}}_N^{\gamma_i}(\boldsymbol{x}; \boldsymbol{\theta}_N^{\gamma_i})$ and $\tilde{\boldsymbol{p}}_V^{\gamma_1}(\boldsymbol{x}; \boldsymbol{\theta}_V^{\gamma_1})$ denote the general solutions $(N^{\gamma_i}, \boldsymbol{p}_N^{\gamma_i}, \boldsymbol{q}_N^{\gamma_i})$ and $(V^{\gamma_1}, \boldsymbol{p}_V^{\gamma_1})$ of BCs shown in Equation (5) at γ_i and Equation (8) at γ_1, respectively. Thus, the derived general solutions can be rewritten as follows

$$\begin{aligned} \tilde{\boldsymbol{p}}_N^{\gamma_i}(\boldsymbol{x}; \boldsymbol{\theta}_N^{\gamma_i}) &= \boldsymbol{B}_N(\boldsymbol{x}) \mathrm{NN}_N^{\gamma_i}(\boldsymbol{x}; \boldsymbol{\theta}_N^{\gamma_i}), \\ \tilde{\boldsymbol{p}}_V^{\gamma_1}(\boldsymbol{x}; \boldsymbol{\theta}_V^{\gamma_1}) &= \boldsymbol{B}_V(\boldsymbol{x}) \mathrm{NN}_V^{\gamma_1}(\boldsymbol{x}; \boldsymbol{\theta}_V^{\gamma_1}), \end{aligned} \tag{17}$$

where $\boldsymbol{B}_N(\boldsymbol{x})$ and $\boldsymbol{B}_V(\boldsymbol{x})$ can be calculated as the unified form $\boldsymbol{I}_{d'+1} - \tilde{\boldsymbol{n}}(\boldsymbol{x})\tilde{\boldsymbol{n}}(\boldsymbol{x})^\top$. It should be noted that d' is set to be 4 for computing $\boldsymbol{B}_N(\boldsymbol{x})$, and d' is equal to 1 for $\boldsymbol{B}_V(\boldsymbol{x})$. $\mathrm{NN}_N^{\gamma_1}$, $\mathrm{NN}_N^{\gamma_2}$ and $\mathrm{NN}_V^{\gamma_1}$ are sub-networks with trainable parameter $\boldsymbol{\theta}_N^{\gamma_i}$ and $\boldsymbol{\theta}_V^{\gamma_1}$, respectively.

Considering that $\mathbf{A_{1r}}$ is far less than 1, we may omit $\mathbf{A_{1r}}$ for accelerating calculation speed. We denote the model without $\mathbf{A_{1r}}$ as $\mathbf{HCPINN-1}$, while denote the model with $\mathbf{A_{1r}}$ as $\mathbf{HCPINN-2}$. We combine all the general solutions on the boundary conditions and construct the ansatz as follows

$$\begin{aligned} (N, \boldsymbol{p}_N, \boldsymbol{q}_N) &= l^{\partial\Omega}(\boldsymbol{x}) \mathrm{NN}^{\mathrm{main}}(\boldsymbol{x}; \boldsymbol{\theta}^{\mathrm{main}})[1:5] \\ &\quad + \sum_{i=1}^{2} (1 - l^{\gamma_i}(\boldsymbol{x})) \tilde{\boldsymbol{p}}_N^{\gamma_i}(\boldsymbol{x}; \boldsymbol{\theta}_N^{\gamma_i}), \\ (V, \boldsymbol{p}_V) &= l^{\partial\Omega}(\boldsymbol{x}) \mathrm{NN}^{\mathrm{main}}(\boldsymbol{x}; \boldsymbol{\theta}^{\mathrm{main}})[6:8] \\ &\quad + (1 - l^{\gamma_1}(\boldsymbol{x})) \tilde{\boldsymbol{p}}_V^{\gamma_1}(\boldsymbol{x}; \boldsymbol{\theta}_V^{\gamma_1}), \end{aligned} \tag{18}$$

where $\mathrm{NN}^{\mathrm{main}}$ is the main neural network with trainable parameters $\boldsymbol{\theta}^{\mathrm{main}}$, $[1:5]$ and $[6:8]$ means taking the first five elements and the last three elements of the output of $\mathrm{NN}^{\mathrm{main}}(x)$ respectively. In Equation (18), we utilize extended distance functions $l^{\partial\Omega}$ and l^{γ_i} to "divide" the problem domain into several parts, where $\tilde{\boldsymbol{p}}_N^{\gamma_i}(\boldsymbol{x}; \boldsymbol{\theta}_N^{\gamma_i})$ and $\tilde{\boldsymbol{p}}_V^{\gamma_1}(\boldsymbol{x}; \boldsymbol{\theta}_V^{\gamma_1})$ are responsible for the approximation on the boundaries while $\mathrm{NN}^{\mathrm{main}}$ is responsible for the internal. Given that Equation (4) is time-dependent, we embed the initial boundary $N(\boldsymbol{x}, 0) = N_0$ into the following ansatz

$$\hat{N} = N(\boldsymbol{x}, t)(1 - \exp(-\beta_t t)) + N_0 \exp(-\beta_t t), \tag{19}$$

where $\beta_t \in \mathbb{R}$ is a hyper-parameter of the "hardness" in the temporal domain. The proposed neural network is trained by minimizing the loss function $\mathcal{L} = \mathcal{L}_{pde} + \mathcal{L}_{data}$, where $\mathcal{L}_{pde} = \frac{1}{K}\sum_{i=1}^{K}\left|\frac{\partial N}{\partial t} - \nabla \cdot (-D\boldsymbol{p}_N + \mathbf{A_{1r}}DN + \mathbf{A_{1z}}DN + A_2 D\boldsymbol{q}_N)\right|^2 + \frac{1}{K}\sum_{i=1}^{K}\left|\nabla \cdot (-\sigma \boldsymbol{p}_V)\right|^2 + \frac{1}{K}\sum_{i=1}^{K}(\left|\boldsymbol{p}_N - \nabla N\right|^2 + \left|\boldsymbol{q}_N - N\nabla N\right|^2 + \left|\boldsymbol{p}_V - \nabla V\right|^2)$, and $\mathcal{L}_{data} = \frac{1}{K_{data}}\sum_{i=1}^{K_{data}}(|N - N_d|^2 + |V - V_d|^2)$ in which N_d and V_d are exact values of N and V, respectively. It is noted that since the current density $\mathbf{j}$ is computed by the gradients of the electric potential V, we cannot directly embed Dirichlet BCs shown in Equation (7) into the ansatz , which make it necessary to add $\mathcal{L}_{data}$ for satisfying Dirichlet BCs. After training process, we can obtain the approximation of the ground truth solution to PDEs governing EM process in TSV. We summarize the proposed hard constraint based physics learning method for transient TSV electromigration analysis in Figure 3.

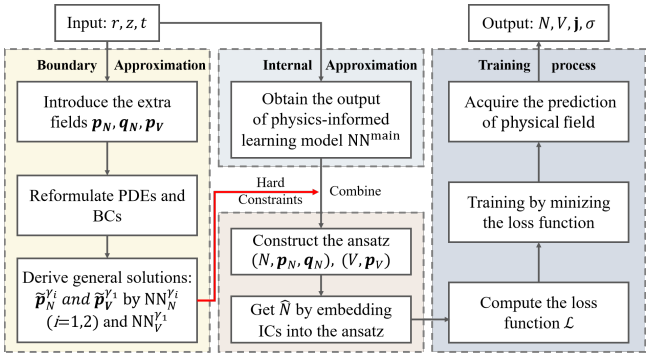


Figure 3: Flow of the proposed hard constraint based physics learning method for transient TSV electromigration analysis.

3.4 Theoretical Analysis

We now present detailed theoretical analysis on the impact of the "*extra fields*" and reformulation. For simplicity, we consider the following abstracted operator equation:

$$\mathcal{F}[u, \frac{\mathrm{d}u}{\mathrm{d}x}, \ldots, \frac{\mathrm{d}^n u}{\mathrm{d}x^n}, \frac{\mathrm{d}}{\mathrm{d}x}(u\frac{\mathrm{d}u}{\mathrm{d}x}), \ldots, \frac{\mathrm{d}^n}{\mathrm{d}x^n}(u\frac{\mathrm{d}u}{\mathrm{d}x})] = f(x), \tag{20}$$

where $f(x)(x \in \Omega)$ is a source function. Let $p = \frac{du}{dx}$, $q = u\frac{du}{dx}$, and we then reformulate the operator equation (20) as follows:

$$\mathcal{F}[u, p, \frac{\mathrm{d}p}{\mathrm{d}x}, \ldots, \frac{\mathrm{d}^{n-1}p}{\mathrm{d}x^{n-1}}, q, \frac{\mathrm{d}q}{\mathrm{d}x}, \ldots, \frac{\mathrm{d}^{n-1}q}{\mathrm{d}x^{n-1}}] = f(x). \tag{21}$$

It can be seen from (20) and (21) that the orders of derivatives in the operator equation are reduced, which can lead to less accumulation of back-propagation, and thus stabilize the training process. To formally and generally explain this mechanism, we consider a simple case as follows

$$\begin{aligned} \frac{\mathrm{d}^2 u}{\mathrm{d}x^2} + \frac{\mathrm{d}}{\mathrm{d}x}(u\frac{\mathrm{d}u}{\mathrm{d}x}) &= -a^2 \sin(ax) + a^2 \cos(2ax),\ x \in (0, 2\pi) \\ u(x) &= 0,\ x = 0 \cup x = 2\pi. \end{aligned} \tag{22}$$

The corresponding ansatz employs a single-layer neural network of width K, where $\hat{u} = \boldsymbol{c}^T\sigma(\boldsymbol{\omega}x + \boldsymbol{b})$, $\hat{p} = \boldsymbol{c}_p^T\sigma(\boldsymbol{\omega}x + \boldsymbol{b})$, $\hat{q} = \boldsymbol{c}_q^T\sigma(\boldsymbol{\omega}x + \boldsymbol{b})$, $\boldsymbol{c}, \boldsymbol{\omega}, \boldsymbol{b}, \boldsymbol{c}_p, \boldsymbol{c}_q \in \mathbb{R}^K$, and σ is an element-wise activation function which we assume to be tanh for simplicity. Considering that the BCs in this case are Dirichlet BCs and the gradients of loss term associated with Dirichlet BCs remain unchanged, we focus on the the gradients of the PDEs loss terms and propose the following theorem

Theorem 3. *Let $\boldsymbol{\theta} = (\boldsymbol{c}, \boldsymbol{w}, \boldsymbol{b})$, $\tilde{\boldsymbol{\theta}} = (\boldsymbol{c}, \boldsymbol{w}, \boldsymbol{b}, \boldsymbol{c}_p, \boldsymbol{c}_q)$, and $\mathcal{L}_{\mathcal{F}}$ as well as $\tilde{\mathcal{L}}_{\mathcal{F}}$ be the loss terms of the original and reformulated PDEs, respectively. The bounds for the gradients of the loss terms are given by*

$$\begin{aligned} |(\nabla_{\boldsymbol{\theta}}\mathcal{L}_{\mathcal{F}})^\top| =& O(\max(|\boldsymbol{c}|^\top, (\boldsymbol{c}^\top)^2)\boldsymbol{w}^2 + a^2) \cdot (\max(1, |\boldsymbol{c}|^\top)\boldsymbol{w}^2, \\ & \max(|\boldsymbol{c}|, \boldsymbol{c}^2) \circ |\boldsymbol{w}| \circ (|\boldsymbol{w}| + 1), \max(|\boldsymbol{c}|, \boldsymbol{c}^2) \circ \boldsymbol{w}^2) \\ |(\nabla_{\tilde{\boldsymbol{\theta}}}\tilde{\mathcal{L}}_{\mathcal{F}})^\top| =& O(\|\boldsymbol{c}_p\|_1 + \|\boldsymbol{c}_q\|_1 + \max(|\boldsymbol{c}|^\top, |\boldsymbol{c}_p|^\top, |\boldsymbol{c}_q|^\top, (\boldsymbol{c}^\top)^2) \\ & |\boldsymbol{w}| + a^2) \cdot (\max(1, |\boldsymbol{c}|^\top)|\boldsymbol{w}|, \max(|\boldsymbol{c}|, |\boldsymbol{c}_p|, |\boldsymbol{c}_q|, \\ & |\boldsymbol{c}^2|) \circ \max(|\boldsymbol{w}|, \boldsymbol{1}), \max(|\boldsymbol{c}|, |\boldsymbol{c}_p|, |\boldsymbol{c}_q|, |\boldsymbol{c}^2|) \circ \max \\ & (|\boldsymbol{w}|, \boldsymbol{1}), \max(|\boldsymbol{w}|, \boldsymbol{1}), \max(|\boldsymbol{w}|, \boldsymbol{1})) \end{aligned} \tag{23}$$

Table 1: The Utilized Parameters for Modeling.

Parameter	Value	Description
N_0	1.53e46Atoms/m^3	Initial atomic concentration
E_A	2.1 eV	Activation energy
k	1.38e-23 J/K	Boltzmann constant
e	1.6e-19 C	Electric charge
Z^*	4	Effective valence charge
ρ_0	1.8e-8 Ω·m	Electrical resistivity
D_0	1e-7 m^2	Self-diffusion coefficient
Q^*	1.387e-20	Heat of transport
Ω	1.182e-29 m^3	Atomic volume
B	5e7 Pa	Back flow stress modular

where $\circ$, $\max(\cdot)$, *and operations on vectors are the element-wise multiplication, the element-wise maximum of vectors, and element-wise operations respectively.* $\mathbf{1}$ *is an all-ones vector.*

The proof of Theorem 3 is not shown due to length limit of paper. From Equation (23), it is observed that $(\nabla_{\boldsymbol{\theta}}\mathcal{L}_{\mathcal{F}})^\top$ has a higher order relationship with the trainable parameters $\boldsymbol{\theta}$ compared with $(\nabla_{\tilde{\boldsymbol{\theta}}}\tilde{\mathcal{L}}_{\mathcal{F}})^\top$. This relationship often leads to training instability, as the gradients $(\nabla_{\boldsymbol{\theta}}\mathcal{L}_{\mathcal{F}})^\top$ may rapidly explode or vanish when $\boldsymbol{\theta}$ are large or small. The stability enhancement of reformulated PDEs arises from lower orders of gradients, independent of the specific form of the PDEs.

4 RESULTS AND DISCUSSIONS

All experiments are implemented in Python 3.9.15 with PyTorch 1.13.1 and are carried out on a Linux server equipped with 2.20-GHz Xeon processors and an NVIDIA RTX A6000. The ground truth data is obtained from the finite element simulations performed by the COMSOL Multiphysics software. The parameters utilized for modeling are listed in Table 1. During the training process, Adam optimization is initially performed, followed by L-BFGS for the subsequent training stage. The initial learning rate of Adam is set to be 0.001, and the activation function is tanh. The scaling configurations $r/10^{-6}$, $z/10^{-6}$, $t/10^{6}$, $D/10^{-18}$, $N/10^{46}$, $V/10^{-2}$, $\sigma/10^{6}$, $\mathbf{j_r}/10^{10}$, and $\mathbf{j_z}/10^{10}$ are employed to scale the inputs and coefficients during the training.

4.1 Accuracy Analysis

The accuracy between the proposed HCPINN method and the finite element method is measured by the relative L2 error. We denote the exact value as $u(r_i, z_i, t_i)$ derived by the finite element method and its approximation as $\hat{u}(r_i, z_i, t_i)$ calculated by the neural network method. For the data points $\{r_i, z_i, t_i\}_{i=1}^M$, the relative L2 error is defined as follows:

$$\text{L2error} = \frac{\sqrt{\sum_{i=1}^{M} |\hat{u}(r_i, z_i, t_i) - u(r_i, z_i, t_i)|^2}}{\sqrt{\sum_{i=1}^{M} |u(r_i, z_i, t_i)|^2}}. \tag{24}$$

We first examine the prediction accuracy of the proposed method for calculating atomic concentration N, electric potential V, current density $\mathbf{j}$ and back flow stress σ. It can be observed from Figure 4 that the bottom has depleted atoms ($\Delta N/N < 0$), and the top has accumulated atoms ($\Delta N/N > 0$) due to electromigration. The

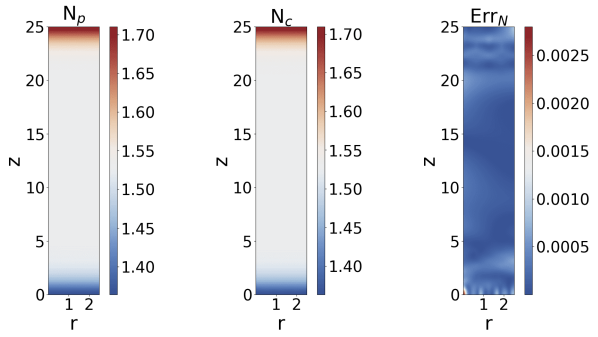


Figure 4: The scaled atomic concentration at t=5e6s: $\mathrm{N_p}$ **and** $\mathrm{N_c}$ **are obtained by the proposed method and the finite element method, respectively, and** $\mathrm{Err_N}$ **is the absolute error between** $\mathrm{N_p}$ **and** $\mathrm{N_c}$**.**

results acquired by the proposed method fit well with the solutions obtained from FEM, with L2error less than 0.25%. As can be seen from Figure 5, the scaled electrical potential of top electrodes is 1.2V and the bottom of TSV is ground. Compared with FEM, the maximum L2error of the proposed model is about 0.3%, exhibiting the high prediction accuracy of calculating electrical potential V. The experimental results regarding current density $\mathbf{j}$, as shown in Figure 6, indicate that the current density near the bottom changes rapidly because the atom depletes at the bottom and then the resistivity of the bottom changes accordingly. Figure 7 shows the prediction of the back flow stress, which arises from the compression at atomic depletion as well as tension at atomic accumulation, and exhibits a lasting impact on atomic movement caused by electromigration. We can find from Figure 7 that compressive stress is present at the top and tensile stress at the bottom, featuring a significant stress gradient throughout TSV.

Table 2 provides the relative L2 errors of multi-physics variables N, V, $\mathbf{j}$, and σ during the time range $0 - 5 \times 10^6$s at the boundary $r = R_{tsv}$, $z = 0$ and $z = h$ calculated by PINN, the proposed models HCPINN-1 and HCPINN-2, respectively. It should be noted that the relative L2 error at the boundary $z = 0$ is not available since the electrical potential at this boundary is zero. From Table 2, we can observe that the errors of the proposed models decrease significantly compared to the PINN method. The error at the boundary $r = R_{tsv}$ is less than those at other boundaries and the error at the boundary $z = 0$ decreases significantly faster in comparison with those at the boundaries $r = R_{tsv}$ and $z = h$.

4.2 Performance Analysis

For verifying the effectiveness of our approach, we compare the proposed models HCPINN-1 and HCPINN-2 with PINN [9], PFNN [32], PINN-LA [15] as well as PINN-LA-2 [15], and LBPINN [33]. It can be seen from Table 3 that the proposed physics-informed neural network models have higher accuracy compared with other PINN-based methods. The prediction accuracy of $N, V, \mathbf{j}, \sigma$ obtained from HCPINN-1 is similar to that from HCPINN-2. Figure 8 shows the loss values $\mathcal{L}_{data}$ of the proposed models HCPINN-1 and HCPINN-2 and other competing PINN based methods over iteration in the training

Table 2: Relative L2 Error (%) comparison between PINN and the proposed models HCPINN-1 and HCPINN-2.

Physical field	Error at $r = R_{tsv}$			Error at $z = 0$			Error at $z = h$		
	PINN	HCPINN-1	HCPINN-2	PINN	HCPINN-1	HCPINN-2	PINN	HCPINN-1	HCPINN-2
N	0.119	0.0205	**0.0196**	0.399	0.0472	**0.0375**	0.390	**0.0197**	0.0333
V	0.108	**0.0423**	0.0457	NA	NA	NA	0.0151	0.0126	**0.0089**
j	2.91	1.20	**0.898**	18.7	4.82	**3.43**	1.37	0.672	**0.445**
σ	6.12	1.06	**1.01**	4.88	0.576	**0.458**	5.14	**0.259**	0.438

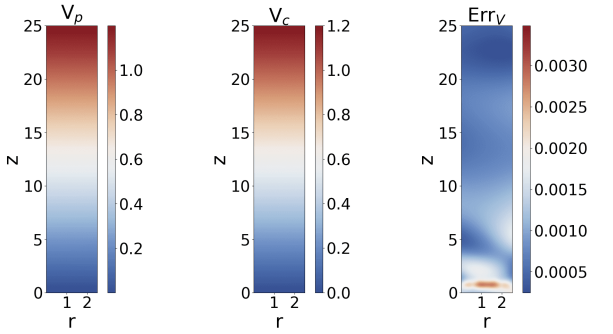


Figure 5: The scaled electrical potential at t=5e6s: V_p and V_c are obtained by the proposed method and the finite element method, respectively, and Err_V is the absolute error between V_p and V_c.

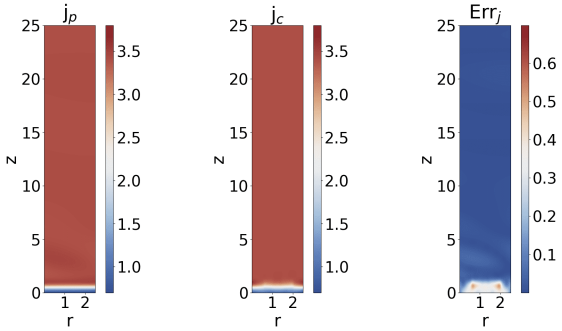


Figure 6: The scaled electric current density at t=5e6s: j_p and j_c are obtained by the proposed method and the finite element method, respectively, and Err_j is the absolute error between j_p and j_c.

process, illustrating that the proposed approach can achieve state-of-art performance as the proposed models have the minimum loss values.

Although it takes much time in the process of training, the inference time of the proposed physics-informed learning model for generating more results with same mesh density is much shorter compared with FEM. It can be seen from Table 4 that the inference speed of the proposed HCPINN method is much faster than FEM and the acceleration ratio can reach to about 661X. Although the acceleration ratio drops when the mesh gets denser, the acceleration ratio can still reach to about 101X when the mesh density gets larger. It is also worthy to mention that the mean squared error

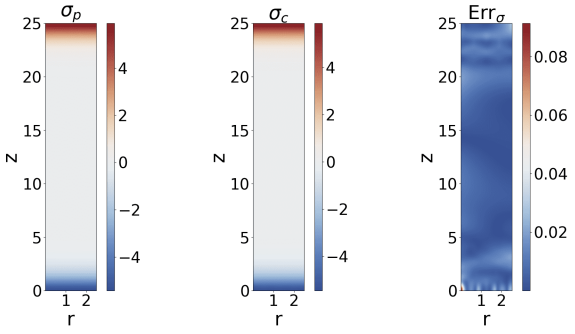


Figure 7: The scaled back flow stress at t=5e6s: σ_p and σ_c are obtained by the proposed method and the finite element method, respectively, and Err_σ is the absolute error between σ_p and σ_c.

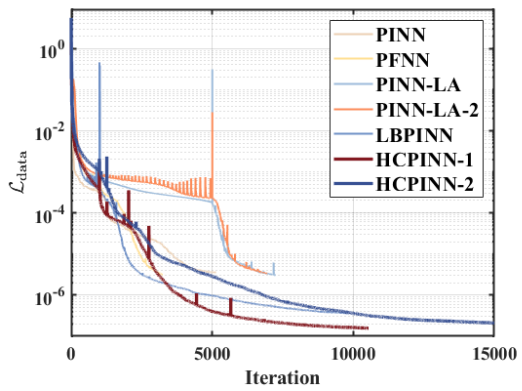


Figure 8: The loss values $\mathcal{L}_{data}$ of the proposed models and other competing PINN based methods over iteration in the training process.

of atomic concentration N decreases when the mesh density gets larger, which shows the potential of the proposed PINN model for EM analysis of TSV.

4.3 Ablation Study

In this section, we analyse the impacts of the "$extra\ fields$" and the "hardness" hyper-parameter β_t in Equation (19). We first consider Equation (22) with $a = 2$, and train two PINN models with and without the "$extra\ fields$", in which there are $N_f = 128$ collocation points and $N_b = 2$ boundary points for 10000 Adam iterations with

Table 3: Relative L2 error (%) comparison with PINN-based Methods during the time range $0 - 5 \times 10^6$s.

Physical field	PINN[9]	PFNN[32]	PINN-LA[15]	PINN-LA-2[15]	LBPINN[33]	HCPINN-1	HCPINN-2
N	0.231	0.122	0.125	0.192	0.052	**0.0151**	**0.0151**
V	0.262	0.119	0.106	0.082	0.082	**0.0409**	**0.0367**
j	6.69	3.72	2.92	1.49	1.58	**0.745**	**0.733**
σ	11.89	6.29	6.44	6.14	2.68	**0.779**	**0.777**

Table 4: The inference time of the proposed PINN model and the computation time of FEM.

Mesh Density	Proposed Model(s)	FEM(s)	MSE of N	Acceleration Ratio
40000	0.1268	25	5.69×10^{-5}	197.21
80000	0.1278	51	3.63×10^{-5}	399.1
160000	0.1633	108	2.27×10^{-5}	661.4
320000	0.3881	229	1.39×10^{-5}	590.1
640000	4.867	493	8.45×10^{-6}	101.2

Table 5: Relative L2 errors (%) of physical field solutions for different hyper-parameter values over the time range $0 - 5 \times 10^6$s.

Term	$\beta_t = 1$	$\beta_t = 2$	$\beta_t = 5$	$\beta_t = 8$	$\beta_t = 10$	$\beta_t = 20$
N	0.0282	0.0262	0.0210	0.0196	0.0173	**0.0148**
V	0.0512	0.0465	0.0471	0.0489	0.0479	**0.0436**
j	0.6529	0.6567	0.7084	0.7936	0.6145	**0.6044**
σ	1.4532	1.3496	1.0822	1.0117	0.8893	**0.7648**

the learning rate 0.001. We test different network architectures including $[1]+3\times[50]+[\cdot]$, $[1]+3\times[100]+[\cdot]$, $[1]+5\times[50]+[\cdot]$, and $[1]+5\times[100]+[\cdot]$. The numbers of outputs are set to be 1 and 3 for the PINN models without and with the "*extra fields*", respectively. In Figure 9, we show the ratio of the moving variance (Movvar) of $\left(\left|\overline{\nabla_\theta \mathcal{L}_\mathcal{F}}\right|\right)$ to that of $\left(\left|\overline{\nabla_{\tilde{\theta}} \tilde{\mathcal{L}}_\mathcal{F}}\right|\right)$ at each iteration. The Movvar of $\left(\left|\overline{\nabla_\theta \mathcal{L}_\mathcal{F}}\right|\right)$ and the Movvar of $\left(\left|\overline{\nabla_{\tilde{\theta}} \tilde{\mathcal{L}}_\mathcal{F}}\right|\right)$ are similar to variables defined in [27]. It can be observed from Figure 9 that the PINN models with the "*extra fields*" can reduce oscillations of gradients during optimization, which is consistent with our theoretical analysis introduced in Section 3.4.

We conduct the experiments with different "hardness" hyper-parameter β_t while all other settings stay the same. Table 5 shows the relative L2 error of multi-physical field solutions over the time range $0 - 5 \times 10^6$s. It can be observed from Table 5 that the accuracy improves with the value of β_t increasing. The reason is that the "hardness" hyper-parameter β_t improves compliance with initial conditions, which in turn enhances the accuracy of the approximations under physical constraints. However, large β_t will lead to performance degradation, which can be seen from Equation (19).

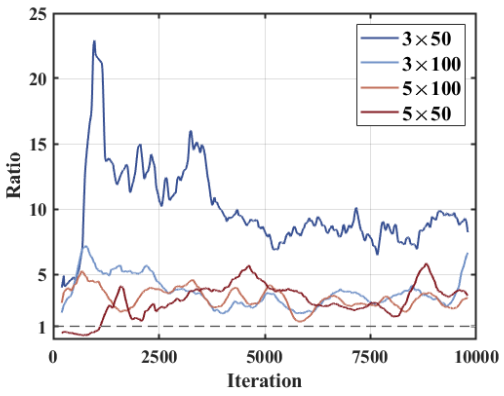


Figure 9: The ratio of MovVar $\left(\left|\overline{\nabla_\theta \mathcal{L}_\mathcal{F}}\right|\right)$ / MovVar $\left(\left|\overline{\nabla_{\tilde{\theta}} \tilde{\mathcal{L}}_\mathcal{F}}\right|\right)$ over iteration in the training process.

5 CONCLUSION

In this paper, we propose a hard constraint based physics-informed neural network method, HCPINN, to calculate the evolution of atomic concentration, stress distribution, and electric current during electromigration process in TSV. To solve PDEs with general BCs, we utilize the "*extra fields*" to reformulate the PDEs and derive the general solutions of BCs in EM analysis of TSV. Besides, we construct the ansatz to enforce hard constraints in physics-informed learning model for EM analysis. The proposed extra field method for deriving the general solution corresponding to more general BCs can effectively stabilize the training process. By enforcing hard constraints into the physics-informed neural network, the proposed HCPINN model can remove the unnecessary loss terms related to BCs and overcome the unbalanced competition between different loss terms, which is more effective in geometrically complex problem domain with multi boundaries. Unlike traditional numerical methods, the hard constraint based physics-informed learning model no longer relies on grid meshing and temporal discretization. Compared with the finite element method, the proposed HCPINN model can lead to about a 100 times speedup with less than 0.1% relative error. Experimental results also demonstrate that the proposed model shows promising accuracy over competing learning-based methods for transient TSV electromigration analysis.

ACKNOWLEDGMENTS

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